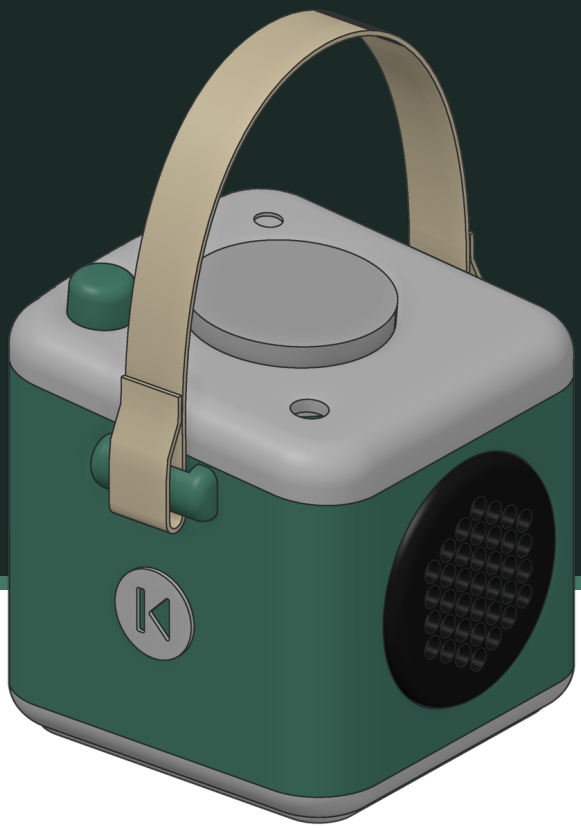
HARDWARE DOCS · BUILD GUIDE

# Timon Box

## Electronics, printing and assembly

Populate the boards, wire them up, print the case and put everything together step by step.



github.com/aaronblankenburg/timon-box
Model: models/box.step
Boards: hardware/ (KiCad 8)
Background: doc/hardware.md
As of: October 2026
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# 1 Overview

The case consists of ten printed parts. Frame, bottom and lid are glued into one shell. At the bottom there is a hatch held by four M3 screws in heat-set inserts. Through it you can still reach the electronics after assembly.

The speaker side is the front. Seen from the front, the “previous” button is on the left and “next” on the right, matching J2 and J3 (chapter 5.4). The lid is tilted forward by 5°, the back with the USB-C socket is slightly curved.

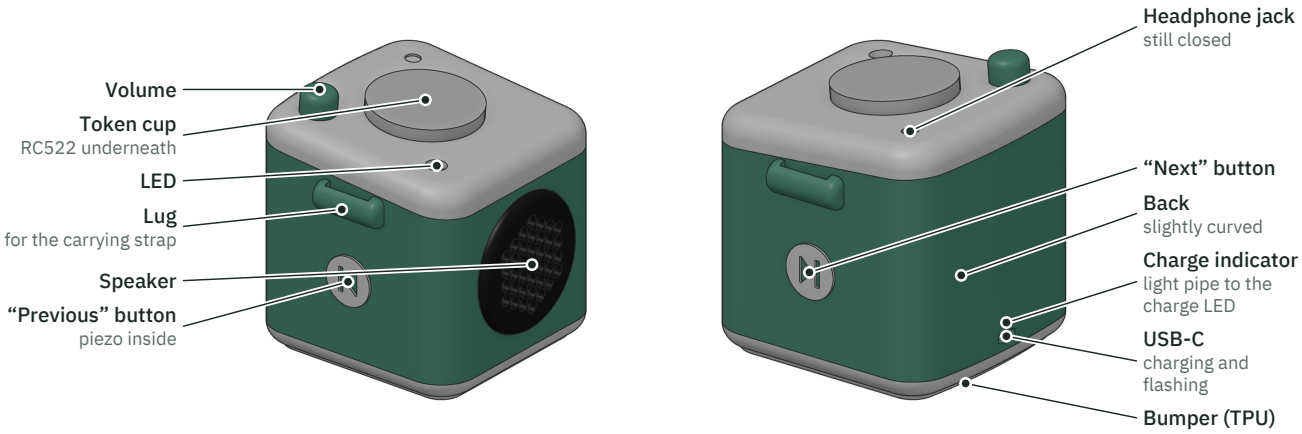



**Fig. 1** *Left from the front, right from the back.*

| Dimension | Value |
|---|---|
| Footprint | 94 × 90 mm, 102 mm wide with lugs |
| Height with bumper | back 94.5 mm, front 87 mm |
| On top | token 5 mm, knob 9 mm above the lid surface |

## At a glance

| PRINTING | JOINING | SCREWS | CELLS |
|---|---|---|---|
| PETG or PLA, bumper TPU, 0.2 mm layer height | Super glue, 2-part epoxy, UV resin, hot glue, tape | 4 × M3 × 5 in M3 heat-set inserts | 2 × 18650, wired in parallel |

> **Note:** Firmware, boards, model and this guide live in the repo github.com/aaronblankenburg/timon-box. Chapter 5 covers the electronics, chapter 6 the assembly. Background on the circuit is in **doc/hardware.md**, the firmware’s pin assignment in **doc/architecture.md**, tuning the knock sensors in **doc/development.md**.

# 2 Printed parts

The part names are the ones from the STEP model `models/box.step`, where all parts sit in their assembled position. The spacer for the electronics is a separate file, `models/pcb_spacer.step`.

| Part | Function | Material | Fixing |
|---|---|---|---|
| `front` | Lid with token cup, LED, potentiometer, headphone jack and pocket for the microswitch | PETG/PLA | Super glue onto `frame` |
| `frame` | Frame: side walls with lugs for the carrying strap, speaker opening, USB-C cutout. At the buttons the wall is thinned to 1 mm on the inside | PETG/PLA | – |
| `back` | Bottom: frame around the hatch with four threaded inserts | PETG/PLA | Super glue to `frame` |
| `door` | Bottom hatch | PETG/PLA | 4 × M3 × 5 into `back` |
| `bumper` | Foot under the hatch | TPU | Super glue to `door` |
| `speaker grill` | Speaker grill, covered with fabric | PETG/PLA | pressed in, super glue |
| `prev button` | “Previous” button, Ø 20 × 1 mm | PETG/PLA | Super glue, outside onto `frame` |
| `next button` | “Next” button, Ø 20 × 1 mm | PETG/PLA | Super glue, outside onto `frame` |
| `volume` | Volume knob | PETG/PLA | pushed onto the potentiometer shaft |
| `token` | Token with a printed-in NFC tag, one per album | PETG/PLA | – |
| `pcb_spacer` | Spacer between main board and RC522, separate file `models/pcb_spacer.step` | PETG/PLA | clipped in |

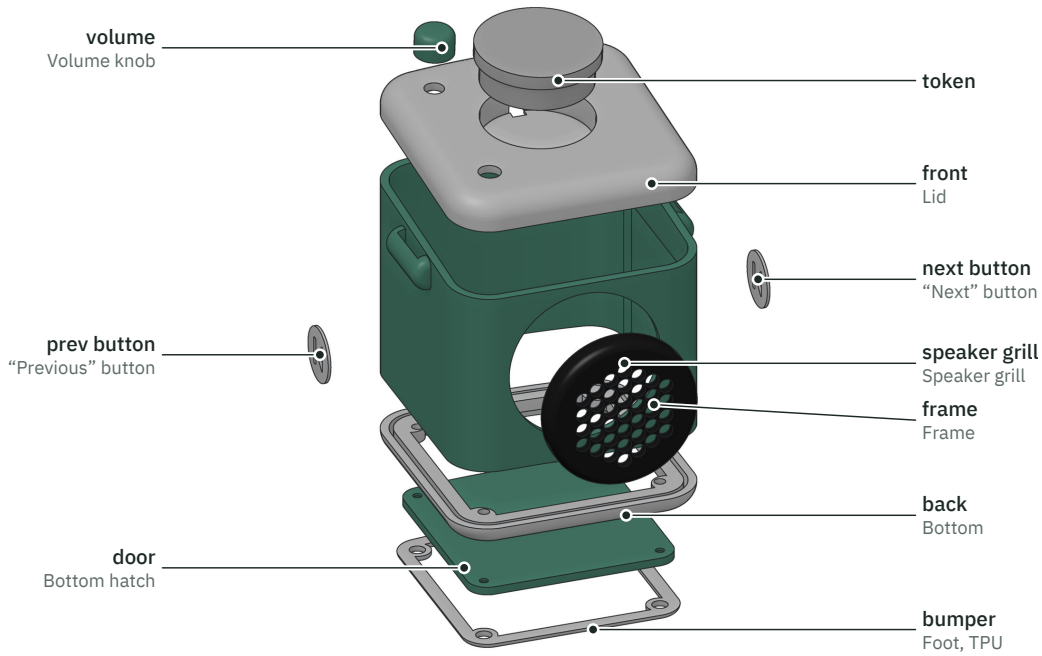



**Fig. 2** *Exploded view. At the bottom the hatch with the bumper, at the sides the buttons, at the front the speaker grill.*

# 3 Printing

All parts except the bumper in PETG or PLA, the bumper in TPU. PETG handles heat better, for example in a car or on a windowsill.

The quickest way is the ready-made plates in `models/plates/`: one 3MF file per colour, parts already arranged, for beds from 220 × 220 mm. Open them in PrusaSlicer as a project (File › Open Project), and layer height, walls, infill, supports and the pause for the tokens are already set. For Bambu Studio there is a project with all five plates, also ready to go. Other slicers only take the layout. Single parts are in `models/stl/` as STL, named as in the table and already turned the way they go on the bed.

Settings: 0.2 mm layer height, 3 walls, 15–20% infill. The NFC pocket in the token is designed for 0.2 mm layers.

| Part | On the bed | Supports | Note |
|---|---|---|---|
| `frame` | Bottom edge | in the speaker hole and under the lugs | The grill has only 0.2 mm clearance: deburr the hole |
| `front` | Bottom edge, top side up | inside under the ceiling, from the build plate only (tree supports) | Clear the pockets for microswitch and jack |
| `back` | Side facing the frame (upside down) | none | |
| `door` | Inner side | none | Counterbores face up |
| `bumper` | Glue side | none | TPU, print slowly |
| `speaker grill` | Outer side | none | |
| `prev button`, `next button` | Visible side | none | 1 mm thin |
| `volume` | Top side | none | |
| `token` | Top side (picture side) | none | Pause before the last layer for the NFC tag, see below |
| `pcb_spacer` | Flat side | none | Separate file, see chapter 5.2 |

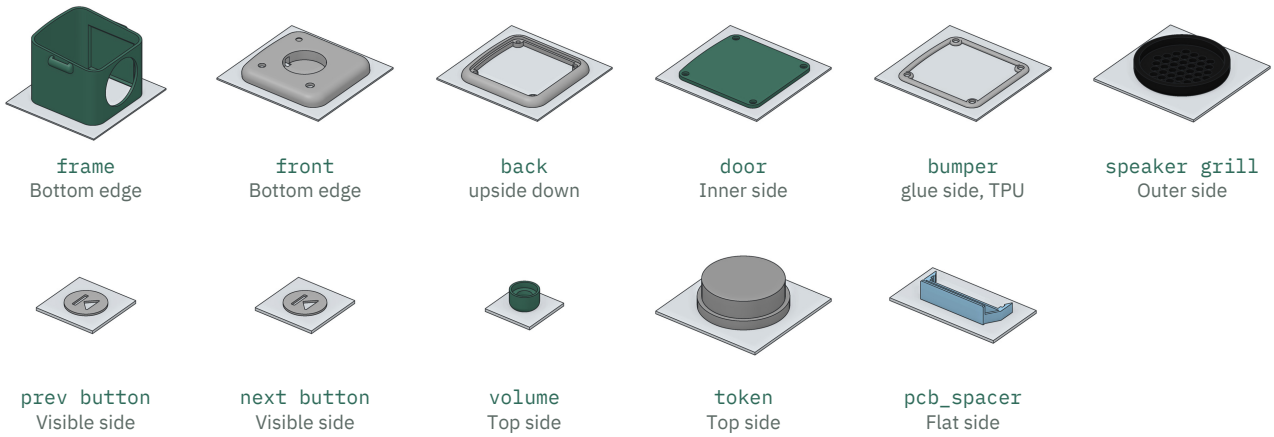



**Fig. 3** *Print orientation, the grey area is the print bed.*

> **Tip:** Print the lid top side up. The other way round, the rounded edges (R 7.5) would sit directly on the bed and come out rough. This way the top and the rounded edges stay smooth, and the supports sit inside where nobody sees them. The hole for the headphone jack is closed at the top by a 0.2 mm layer and stays closed as long as no jack is fitted.

## Token with NFC tag

The token is 15 mm tall and hollow. 0.2 mm above the bottom face there is a pocket (Ø 26 × 0.4 mm) for a round NFC sticker: 25 mm, NTAG, at most 0.4 mm thick.

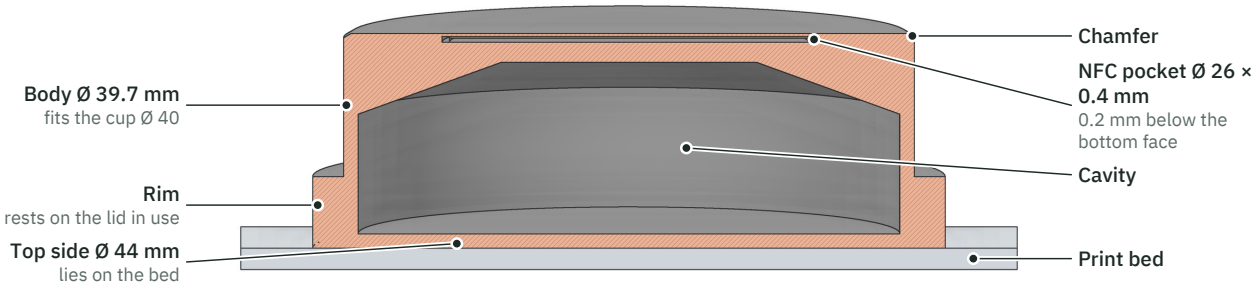
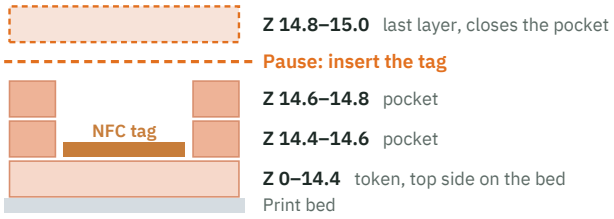



**Fig. 4** *Token in section, the way it is printed.*

- Put the token on the bed with its top side (Ø 44 mm) down. That makes the picture surface smooth, and the wide rim sticks well.
- In the slicer, add a pause before the last layer, with 0.2 mm layers before **Z = 15.0 mm**. The NFC pocket is then open at the top.
- During the pause, put the NFC tag into the pocket, adhesive side down, and press it on. Resume printing, the last layer closes the pocket. The tag then sits 0.2 mm below the bottom face, right above the RC522.
- If the slicer warns about long bridges, that is the ceiling over the cavity in the token. That is expected.
- Roughly cut out the picture from the token sheet (`tools/token-sheet.html`), stick it onto the top side (Ø 44 mm), for example with double-sided tape, and trim it along the edge.




> **Note:** Special tokens for Bluetooth and management are printed the same way. Their UIDs go into `platformio.ini`, see chapter 5.5.

# 4 Materials and tools

## Bought parts for the case

| Part | Qty | Note |
|---|---|---|
| Heat-set insert M3 | 4 | For a Ø 4.0 mm hole, 4 mm long. Up to 5.7 mm works, it then sticks out 1.2 mm on the inside |
| Socket head screw M3 × 5, ISO 4762 | 4 | The Ø 5.5 mm head sinks into the Ø 5.6 mm counterbore and into the bumper. No longer than 5 mm: in two corners a cell sits right above the insert |
| 18650 cell | 2 | Plain cells without protection circuit (65 mm) with solder tabs, same type and same voltage. Wired in parallel |
| Fabric | 1 piece, 8 × 8 cm | Thin and stretchy, for example a sock |
| NFC sticker, round, 25 mm | per token | NTAG, at most 0.4 mm thick |
| Roller shutter strap, about 15 mm wide | about 30 cm | Carrying strap through the lugs, from the hardware store |
| Cable tie, narrow | 1 | Around main board, spacer and RC522 |
| Clear filament, 1.75 mm | a few cm | Light pipe from the charge LED to the outside, may run in a bend |

## Glue and small stuff

| Material | what for |
|---|---|
| Super glue | Case shell, buttons, speaker grill, bumper |
| 2-part epoxy | Speaker, charging board, step-up |
| UV resin with UV lamp | LED, USB-C socket, microswitch |
| Hot glue | Cells |
| Thin double-sided tape | Piezos, RC522, pictures on the tokens |
| Kapton tape, heat-shrink tubing | Insulate cell contacts and solder joints |
| Needle and strong thread or a sewing machine | Carrying strap |
| Sandpaper, primer, paint | optional, for the surface |

## Electronics in the case

The full parts list is in chapter 5.1. These parts have to fit the case:

| Part | Where | Fits if … |
|---|---|---|
| Speaker, 3 W | inside behind the grill | Ø 50–57 mm rests with its rim on the ring (Ø 48.6–54.6 mm); smaller ones sit inside the ring right at the grill |
| Piezo disc, about 35 mm | inside on the side walls | thinned area 56 × 50 mm |
| PL9823, 8 mm | lid, front left | hole Ø 8 mm |
| Potentiometer 10 kΩ | lid, back left | M7 thread (hole Ø 7 mm), round 6 mm shaft. Threaded bushing and nut must fit the 3 mm deep recess in the knob |
| RC522 | under the token cup | cup floor 1 mm |
| Microswitch | pocket next to the cup | lever reaches through the slot into the cup |
| USB-C board | back wall, bottom | cutout 9.2 × 3.5 mm |

| Part | Where | Fits if … |
|---|---|---|
| Power latch with TP4056 | inside of the wall, near the USB-C socket | light pipe from the Ø 2 mm hole to the charge LED |
| Step-up 5 V | free spot on the inside of the wall | not on the thinned button areas |
| Headphone jack 3.5 mm | lid, back right, optional | hole Ø 6 mm, pocket 12 × 7 × 2 mm |

### Tools

Soldering iron (for the inserts a heat-set tip or an old tip), 2.5 mm hex key, UV lamp, hot glue gun, multimeter, saw or side cutter for the potentiometer shaft, utility knife, sandpaper. For the electronics also: soldering iron with a fine tip, solder, flux, desoldering braid, tweezers, wire stripper and a USB-C cable for flashing.

# 5 Electronics

The electronics are two boards and a few loosely wired parts. The assembly drawings come straight from the KiCad files in `hardware/`, the schematics are in the appendix.

- **Main board** (40.9 × 60.5 mm): ESP32, CH340C USB-serial converter, 3.3 V regulator and three modules on female headers: MAX98357A I²S amplifier and SD card on top, RC522 RFID reader underneath.
- **Power latch V2** (31.3 × 18.3 mm): TP4056 charger module, switching between USB and battery, and the self-holding circuit. A microswitch turns the box on when a token is placed. After that the ESP32 keeps the box on by itself. There is no other way to switch it on, not even through USB.
- **Loosely wired:** 18650 cells, USB-C socket, 5 V boost converter, two 2N7000, microswitch, two piezos, potentiometer, LED and speaker.

A 7-pin cable connects the main board to the power side. The USB data lines run through this cable as well, so the USB-C socket on the case charges the battery and is also used for flashing.

## 5.1 Parts list

| Part | Qty | Note |
|---|---|---|
| ESP32-WROOM-32, -32D or -32E | 1 | Without PSRAM. A WROVER does not work, its PSRAM uses GPIO 16 and 17, which the SD card needs |
| MAX98357A module | 1 | I²S amplifier. 7 pins: LRC, BCLK, DIN, GAIN, SD, GND, VIN |
| microSD module for 3.3 V | 1 | 6 pins: 3V3, CS, MOSI, CLK, MISO, GND. The board gets shortened (step E5) |
| RC522 module | 1 | 13.56 MHz RFID. 8 pins, sits under the main board |
| TP4056 charger module with protection | 1 | Gets soldered to the back of the power latch |
| Boost module with fixed 5 V | 1 | Some modules need solder bridges to be set to 5 V |
| 18650 cell | 1–2 | Plain Li-ion cells, 65 mm long. Two in parallel in the printed case. The protection circuit sits on the TP4056 module |
| USB-C socket on a small board | 1 | With 5.1 kΩ on CC1 and CC2, otherwise USB-C chargers deliver no 5 V. D+ and D− must be broken out |
| 2N7000 (TO-92) | 2 | Modification on the power latch (step E8) |
| Microswitch 12.8 × 5.8 mm | 1 | Changeover with C, NO and NC. The token presses it |
| Piezo disc, about 35 mm | 2 | Knock sensors for next and previous |
| Potentiometer 10 kΩ linear | 1 | Volume |
| PL9823, 8 mm | 1 | Through-hole RGB LED with built-in controller, colour order RGB |

| Part | Qty | Note |
|---|---|---|
| Speaker, 3 W | 1 | Small, 4 or 8 Ω |
| JST XH socket, 7 pins, vertical, with plug | 1 | 2.5 mm pitch, for the cable to the power side |
| Female and male headers, 2.54 mm | | 7 pins for the amplifier, 6 pins for the SD card, 8 pins for the RC522 |
| microSD card | 1 | FAT32 |
| NFC stickers, round, 25 mm | per token | 13.56 MHz, NXP NTAG with 7-byte UID. They go into the token during printing (chapter 3) |

## 5.2 Main board

| Part | Value | Package | Job |
|---|---|---|---|
| U1 | ESP32-WROOM-32(D/E) | Module | |
| U2 | AMS1117-3.3 | SOT-223 | 3.3 V from the 5 V on the cable |
| U3 | CH340C | SOIC-16 | USB-serial, no crystal |
| Q1, Q2 | 2N7002 | SOT-23 | Auto reset: DTR and RTS drive EN and IO0 |
| D1, D2 | BZT52C3V3 | SOD-323 | Zener diodes, limit the piezo voltage to 3.3 V |
| R1, R2 | 100 kΩ | 0805 | Load in parallel to the piezos |
| R3 | 10 kΩ | 0805 | Pull-up on EN |
| R4, R5 | 10 kΩ | 0805 | Pull-ups for I²C (spare) |
| C1, C8 | 100 nF | 0805 | Decoupling caps for CH340C and ESP32 |
| C2 | 1 µF | 0805 | With R3 the start-up delay on EN |
| C3 | 10 µF | 0805 | AMS1117 input |
| C4, C7 | 22 µF | 1206 | 3.3 V |
| C5, C6 | – | 0805 | **Do not populate** |
| SW1, SW2 | Push button 6 × 6 mm | PTS645 | RST and BOOT |
| J4 | JST XH, 7 pins, vertical | | Cable to the power side, see step E3 |
| J1, J5 | 3 pins | | Potentiometer and LED, pin header or wires directly |
| J2, J3 | 2 pins | | Piezos, pin header or wires directly |
| J6 | 4 pins | | I²C, spare for extensions. Stays empty |
| MAX1 | Female header, 7 pins | | MAX98357A module |
| U4 | Female header, 6 pins | | SD module |
| P1 | Female header, 8 pins, back side | | RC522 module |

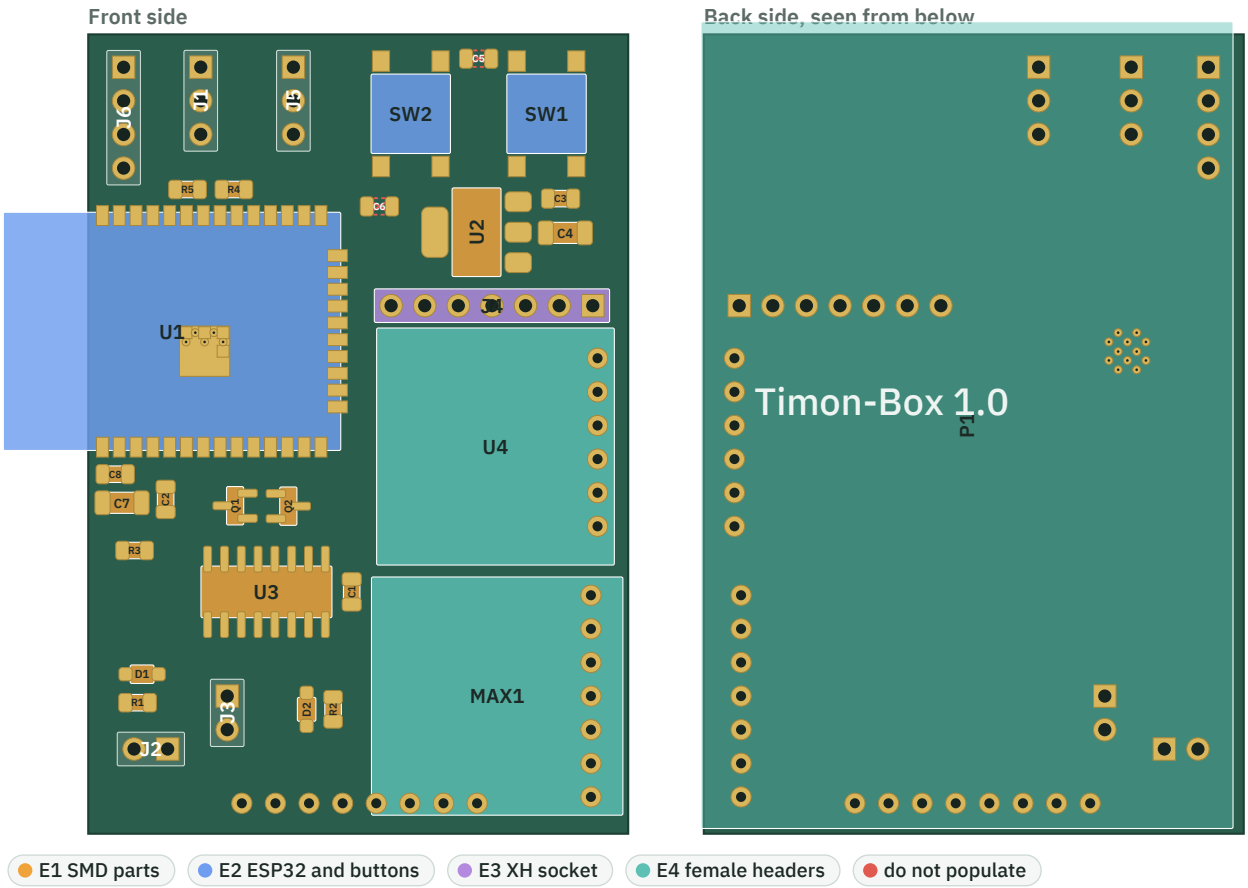



**Fig. 5** *Assembly drawing of the main board from* `hardware/mainboard/timon-box.kicad_pcb`. *The back side is mirrored, as seen from below; the RC522 covers almost all of it.*

## E1 Populate the SMD parts

Soldering iron

- Populate the CH340C (U3), the AMS1117 (U2), both 2N7002 (Q1, Q2), resistors, capacitors and zener diodes. Values in the table, positions in the assembly drawing.
- C5 and C6 stay empty.

## E2 Solder in the ESP32 and the buttons

Soldering iron

- Solder in the ESP32 module (U1) and the two buttons RST (SW1) and BOOT (SW2).
- Only use a WROOM module without PSRAM. On a WROVER the PSRAM uses GPIO 16 and 17, which the SD card needs.

## E3 Fit the XH socket J4

Soldering iron

- The footprint of J4 is a 2.54 mm pin header. What goes in is a JST XH socket with 2.5 mm pitch, it fits the same holes.
- Its housing is deeper than a pin header and hits the AMS1117 (U2). So notch the housing on the underside above the regulator, then solder it in.
- Pin 1 (+5 V) is at the board edge next to C4, pin 7 (D−) towards the ESP32.

## E4 Solder in the female headers

Soldering iron

- On top, solder in the female headers for the amplifier (MAX1, 7 pins) and the SD module (U4, 6 pins), underneath the one for the RC522 (P1, 8 pins).

## E5 Shorten the SD module

- The SD module's board overlaps the XH socket. Saw or sand off the edge facing the socket until the module plugs in next to the socket (photo below).

## E6 Plug in the modules

Soldering iron

- Plug the amplifier and the SD module in on top.
- Solder the pin header to the RC522 so that it sits under the main board.
- Directly opposite the pin header, the printed spacer (`pcb_spacer`) supports the overhanging main board so it does not hang on the pin header alone. The board clips in between its arms with its long edges, its foot stands on the RC522. In the case a cable tie goes around it as well (chapter 6, step 11).
- In the case the RC522 is stuck with its flat back under the token cup (chapter 6, step 9).

> **Note:** On the MAX98357A, GAIN is tied to GND, which gives 12 dB. SD stays open. The firmware sends mono anyway, so the channel does not matter.

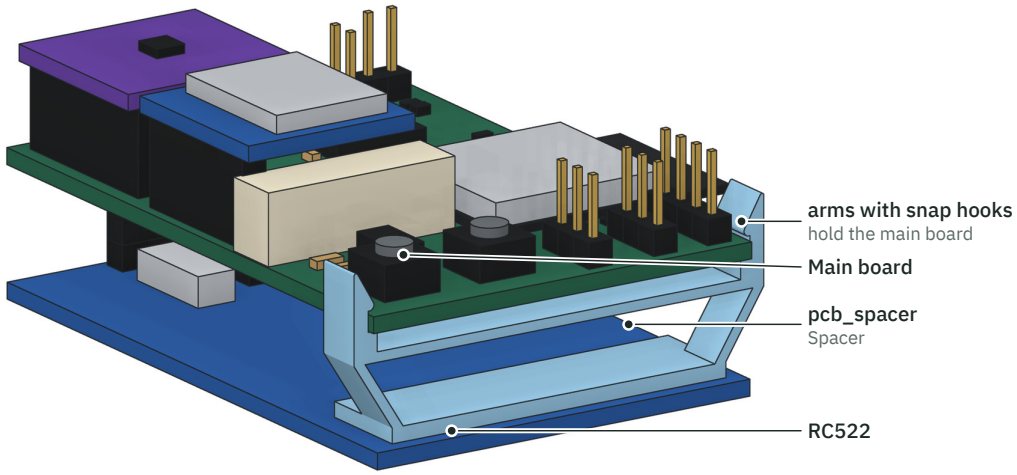
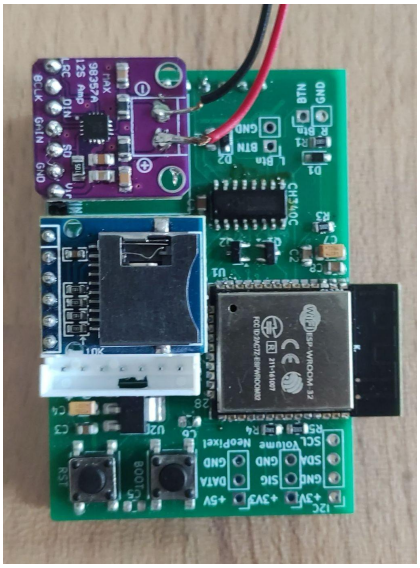
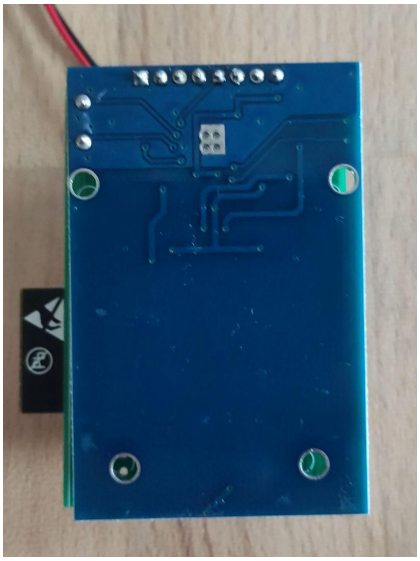


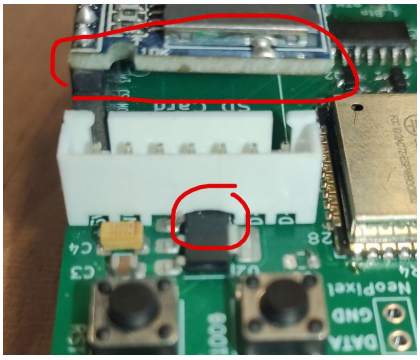

**Fig. 6** *The spacer directly opposite the pin header, between main board and RC522. The pin header is at the other end, at the back of the picture.*



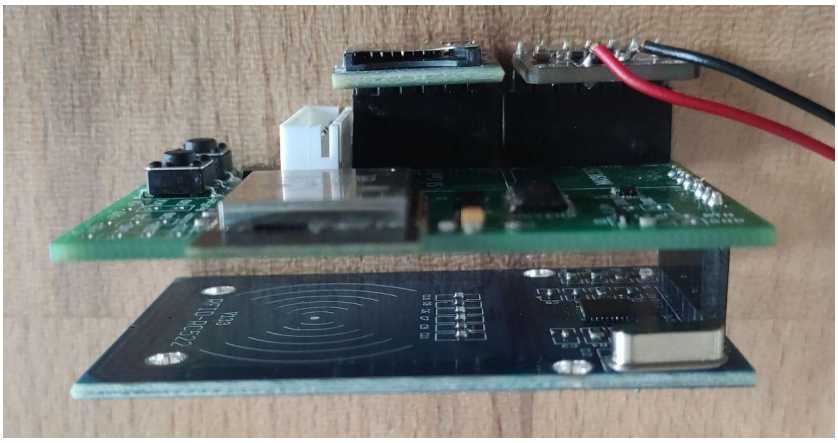
*From above: amplifier (purple), SD module, the XH socket J4 below it, the ESP32 on the right.*



*From below: the RC522.*



*The shortened SD module (circled at the top) and the notched XH socket above the AMS1117 (circled at the bottom).*



*From the side: amplifier and SD module on female headers, the RC522 under the main board.*

## 5.3 Power side

| Part | Value | Package | Job |
|---|---|---|---|
| Q1 | AO4606 | SOIC-8 | N and P channel in one package: P switches Vcc, N holds it |
| Q2 | BSS138 | SOT-23 | TrigOut, open drain to the ESP32 |
| Q3 | BSS138 | SOT-23 | Switch-on pulse |
| Q4 | AO3401A | SOT-23 | Battery branch of the USB/battery switchover |
| D1 | BAT85 | SOD-323 | Clamps the switch-on pulse downwards |
| D2 | BAT85 | DO-35, through-hole | USB branch of the switchover |
| R1, R4–R7 | 10 kΩ | 0805 | R4/R5: voltage divider for Sense |
| R2 | 1 MΩ | 0805 | Pull-down on Hold |
| R3 | 100 kΩ | 0805 | Discharges C1 |
| C1 | 10 µF | 0805 | Switch-on pulse |
| C2 | 1 µF | 0805 | Gate of the P channel |
| J4 | TP4056 module | | Soldered onto the back |
| J1 | – | | The seven wires are soldered in directly here |

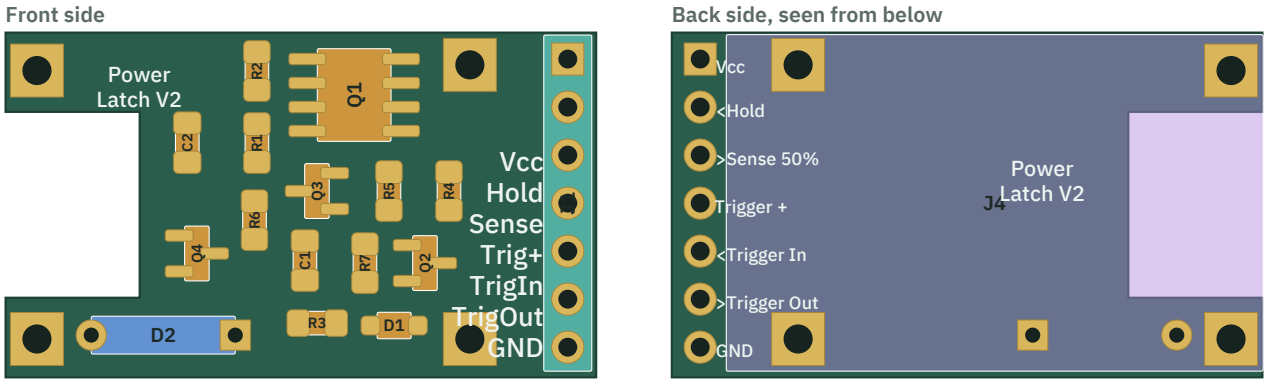



**Fig. 7** *Assembly drawing of the power latch V2 from* `hardware/power-latch/power-latch-v2.kicad_pcb`. *The labels of J1 are on both sides.*

### How the power latch switches

- **USB or battery:** The 5 V from USB reach the load through D2, the battery through Q4. With USB plugged in, Q4 blocks. The box then runs from USB, and the TP4056 charges the battery.
- **Switching on:** A token presses the microswitch, which connects `Trig+` to `TrigIn`. Through C1, Q3 turns on for about a second, long enough for the ESP32 to start and set HOLD.
- **Holding:** While `Hold` is high, the N channel of Q1 keeps the box on. Otherwise R2 pulls `Hold` to GND.
- **Token signal:** While the switch is closed, Q2 pulls `TrigOut` to GND. The ESP32 reads this on GPIO 32.
- **Battery voltage:** R4 and R5 divide `Vcc` by 2 and put the result on `Sense`. The ESP32 measures it on GPIO 39.

### E7 Populate the power latch

Soldering iron

- Populate the SMD parts, the through-hole diode D2 last.
- Then solder the TP4056 module onto the back (J4). Its four corner pads IN+, IN−, OUT+ and OUT− sit directly on the footprint.

## E8 Add two 2N7000

Soldering iron

- `Hold` of the latch is not connected to the ESP32 directly. Two 2N7000 sit in parallel: drain to `Vcc`, source to `Hold`.
- Gate of one (“ESP”) to HOLD from the ESP32, cable pin 3 (blue). Gate of the other (“USB”) to D− of the USB-C socket.
- Together they form a logical OR: `Hold` is high when the ESP32 sets HOLD or when there is voltage on D−. Neither of them can switch the box on: when the box is off, `Vcc` carries no voltage.

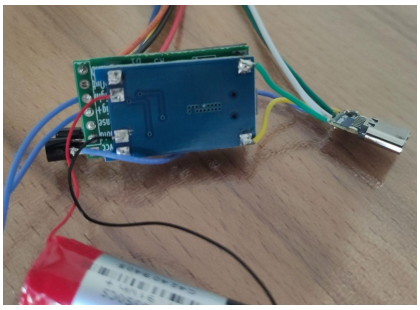
## E9 Check the boost module

- Measure the boost module on its own and without load before connecting it: it has to deliver 5 V. Some modules need solder bridges to be set to 5 V.

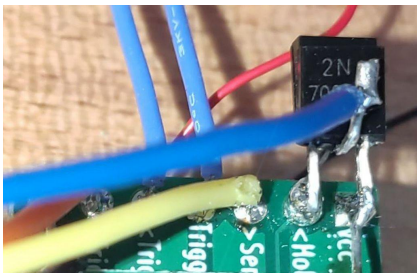
## E10 Wire up the power side

Soldering iron

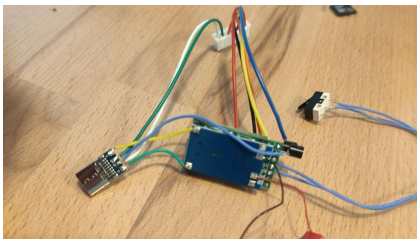
- Solder on the USB-C socket, microswitch, boost module, both 2N7000 and the seven wires of the XH cable following the tables in 5.4. Leave the wires long enough for fitting.
- The NC contact of the microswitch stays free. The cells are added only during assembly (chapter 6, step 11).



*Power latch V2 with the TP4056 module on the back, the USB-C socket on the right, the 18650 cell at the bottom.*



*One of the two 2N7000 on the Hold and Vcc pads of the latch.*



*Power side with microswitch and cable to the main board.*

## 5.4 Wiring

Every row names both ends. The full wiring diagram of the power side is in the appendix.

### Cable between main board and power side

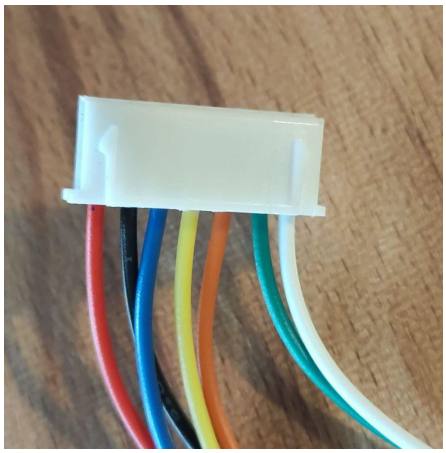
On the main board the cable plugs into J4, on the power side the wires go to different parts. Pin 1 (+5 V) is at the board edge next to C4, pin 7 (D–) towards the ESP32.

| J4 pin | Signal | Wire | ESP32 | Power side |
|---|---|---|---|---|
| 1 | +5V | red | – | Boost OUT+ |
| 2 | GND | black | – | Boost OUT– |
| 3 | HOLD | blue | GPIO 33 | Gate of the 2N7000 “ESP” |
| 4 | SENSE | yellow | GPIO 39 | Latch J1 pin 3, `Sense 50%` |
| 5 | SW_TOKEN | orange | GPIO 32 | Latch J1 pin 6, `Trigger Out` |
| 6 | D+ | green | CH340C | USB-C D+ |
| 7 | D– | white | CH340C | USB-C D– and gate of the 2N7000 “USB” |

### Power latch, connector J1

| Pin | Label | to |
|---|---|---|
| 1 | `Vcc` | Boost IN+ and drain of both 2N7000 |
| 2 | `Hold` | Source of both 2N7000 |
| 3 | `Sense 50%` | Cable pin 4 (yellow) |
| 4 | `Trigger +` | Microswitch C |
| 5 | `Trigger In` | Microswitch NO |
| 6 | `Trigger Out` | Cable pin 5 (orange) |
| 7 | `GND` | Boost IN– |

*The NC contact of the microswitch stays free.*



*The XH plug with its wire colours, pin 1 on the left.*

### Battery, charging and USB-C

| from | to |
|---|---|
| USB-C VBUS, GND | TP4056 IN+, IN–. On the latch these are the pads 5v + and GND |
| 18650 plus, minus (both cells in parallel) | TP4056 B+, B– |
| TP4056 OUT+, OUT– | Latch Bat+, Bat–. The module is soldered on there |
| USB-C D+, D– | Cable pins 6 and 7, D– also to the gate of the 2N7000 “USB” |

### Controls on the main board

| Connector | Pins | GPIO | to |
|---|---|---|---|
| J1 “Volume” | +3V3, SIG, GND | 36 | Potentiometer: ends to +3V3 and GND, wiper to SIG |
| J2 “R Btn” | BTN, GND | 34 | Piezo on the **left** wall, seen from the front: previous |
| J3 “L Btn” | BTN, GND | 35 | Piezo on the **right** wall: next |
| J5 “NeoPixel” | +5V, DATA, GND | 2 | PL9823: VDD, DIN, GND. DOUT stays free |
| J6 „I2C“ | +3V3, GND, SDA, SCL | 14, 13 | Spare, free |
| MAX98357A | +, – | | Speaker |

> **Note:** The labels of J2 and J3 on the board are swapped, and so are they in the firmware. Glue the piezo from J2 to the left wall and you get previous on the left and next on the right.

> **Tip:** The piezos are connected reversed: the brass side (usually the black wire) to BTN, the ceramic side (usually red) to GND. If a piezo only triggers when you let go, swap its wires. If the box gets louder instead of quieter when you turn left, swap the two outer wires on the potentiometer.

## 5.5 First start

Firmware, boards and this guide live in the repo github.com/aaronblankenburg/timon-box. Building and flashing is done with PlatformIO, in VS Code with the PlatformIO extension or with `pio` on the command line. Best do a first start once before fitting, with the electronics lying open, and once more in the case (chapter 6, step 12).

### Per-box values in platformio.ini

Everything that differs from box to box lives in its own env in `platformio.ini`:

```
[env:mybox]
build_flags =
    ${env.build_flags}
    -D BOX_NAME=\"My-Box\"
    -D BOX_COLOR=\"#2f6b5a\"
    -D BLUETOOTH_TAG=\"04c6d4b26e6780\"
    -D MANAGEMENT_SERVER_TAG=\"042041122a6781\"
    -D BATTERY_ADC_LOW=1994
    -D BATTERY_MV_LOW=3685
    -D BATTERY_ADC_HIGH=2390
    -D BATTERY_MV_HIGH=4322
    -D RKS_THRESHOLD=10000
    -D LKS_THRESHOLD=10000
    -D KNOCK_ON_LEVEL=100
    -D KNOCK_MIN_PRESS_MS=25
    -D KNOCK_MAX_GAP_US=700
```

The code only holds fallback values. For a new box, copy an existing env, rename it and adjust the values. Timon's values are a good start.

Strings are in escaped quotes, as with `BOX_NAME`. Flash again after every change.

| Value | Meaning | How to find it |
|---|---|---|
| `BOX_NAME` | Name in Wi-Fi, in Bluetooth and in the web UI | your choice |
| `BOX_COLOR` | Colour of the icon in the browser tab | your choice |
| `BLUETOOTH_TAG`<br>`MANAGEMENT_SERVER_TAG` | UIDs of the two special tokens | Place the token, the console reports `tag: <uid>` (step 4) |
| `BATTERY_ADC_LOW`<br>`BATTERY_MV_LOW`<br>`BATTERY_ADC_HIGH`<br>`BATTERY_MV_HIGH` | Two calibration points for the battery gauge: ADC value and voltage in mV | `battery:` or the management page shows the ADC value, a multimeter the voltage (step 6) |
| `RKS_THRESHOLD`<br>`LKS_THRESHOLD`<br>`KNOCK_…` | Knock sensors | `knockdebug:1` and `knockdebug:2`, see doc/development.md (step 5) |

### Serial console

Plug the USB-C socket into the computer, switch the box on with a token and start `pio device monitor -b 115200`, or the Serial Monitor in VS Code. Type commands with a colon, for example `battery:`, then Enter.

### Step by step

1. **SD card:** FAT32. The root directory needs `tap.mp3` (click when knocking) and `alert.mp3` (warning sound for low battery), both from `sdcard/`. The card layout is described in architecture.md, section 4.
2. **Your own env:** In `platformio.ini`, copy an existing env such as `[env:timon]`, rename it and adjust `BOX_NAME` and `BOX_COLOR`. The other values follow in the next steps.

3. **Flashing:** Switch the box on with the microswitch, USB does not turn it on. Then plug the USB-C socket into the computer and run `~/.platformio/penv/bin/pio run -e <box> -t upload --upload-port /dev/ttyUSBn`.
4. **Special tokens:** One token for Bluetooth mode and one for management mode. Place each of them, the console reports `tag: <uid>`. Enter the UIDs as `BLUETOOTH_TAG` and `MANAGEMENT_SERVER_TAG` in the env and flash again. Every other UID is an album.
5. **Tune the knock sensors:** `knockdebug:1` shows levels and areas every second, `knockdebug:2` also traces the next press. Set the thresholds in the env so that a press triggers reliably and loud music does not (doc/development.md). Flash again.
6. **Calibrate the battery gauge:** ESP32 ADCs vary. The box converts the ADC value to millivolts using two calibration points. Each point is an ADC value plus the voltage a multimeter measures at the same moment between `Vcc` and `GND` on J1 of the latch. On USB, `battery:` shows the ADC value; it and the voltage go into `BATTERY_ADC_HIGH` and `BATTERY_MV_HIGH`. On battery without USB, ideally with a half-empty battery, place the management token: the management page shows the ADC value under “Battery”; it and the voltage go into `BATTERY_ADC_LOW` and `BATTERY_MV_LOW`. Flash again. `battery:` then shows voltage and percentage, the voltage should match the multimeter.

> **Note:** With USB plugged in, the box measures the USB voltage instead of the battery, so the battery gauge shows full. On a USB charger the box also stays on, even if the firmware switches HOLD off.

# 6 Assembly

First the parts are prepared, then frame and bottom are glued together and everything that sits on the walls is fitted. The lid is fitted out separately and glued on at the very end. Until then you can reach everything from above, afterwards only through the bottom hatch.

Colours in the steps: Super glue · 2-part epoxy · UV resin · Hot glue · Double-sided tape

## 6.1 Preparation

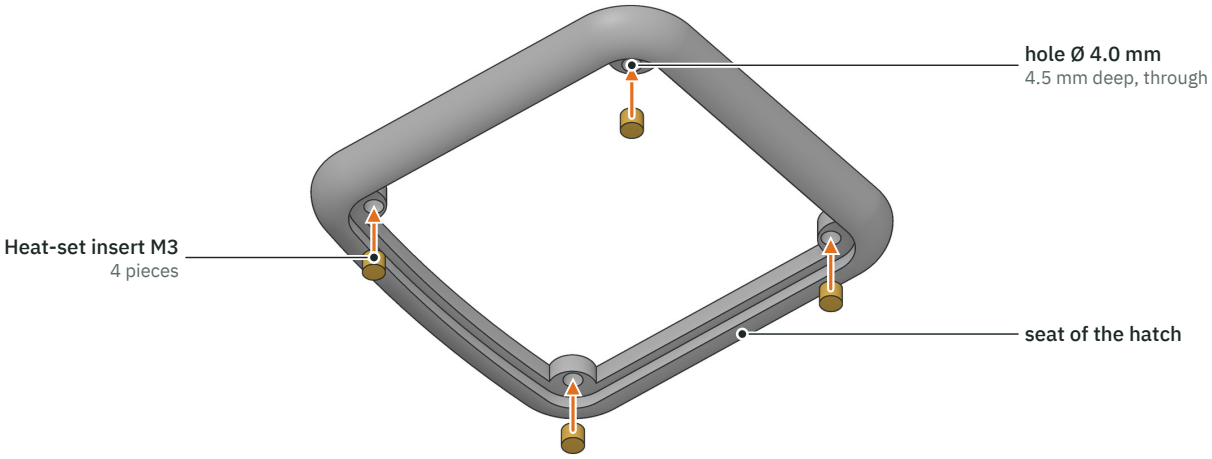



**Fig. 8** `back` *from below, the inserts go in from the hatch side.*

### 1 Prepare the parts

Sanding and paint (optional)

- Remove supports, deburr holes. Test-fit grill, token and hatch: the grill has to go into the opening without force.
- If you want to paint, sand and paint `back`, `frame`, `front` and the buttons now. Mask the glue surfaces and leave them bare, super glue holds better on bare plastic than on paint.

### 2 Melt in the heat-set inserts

Soldering iron

- Put the four M3 inserts into the holes in `back` from below, that is from the side the hatch will rest against later.
- Melt them in straight with the soldering iron until they are flush with the seating surface. Set the iron to roughly the printing temperature of the material.

## 6.2 Lower part

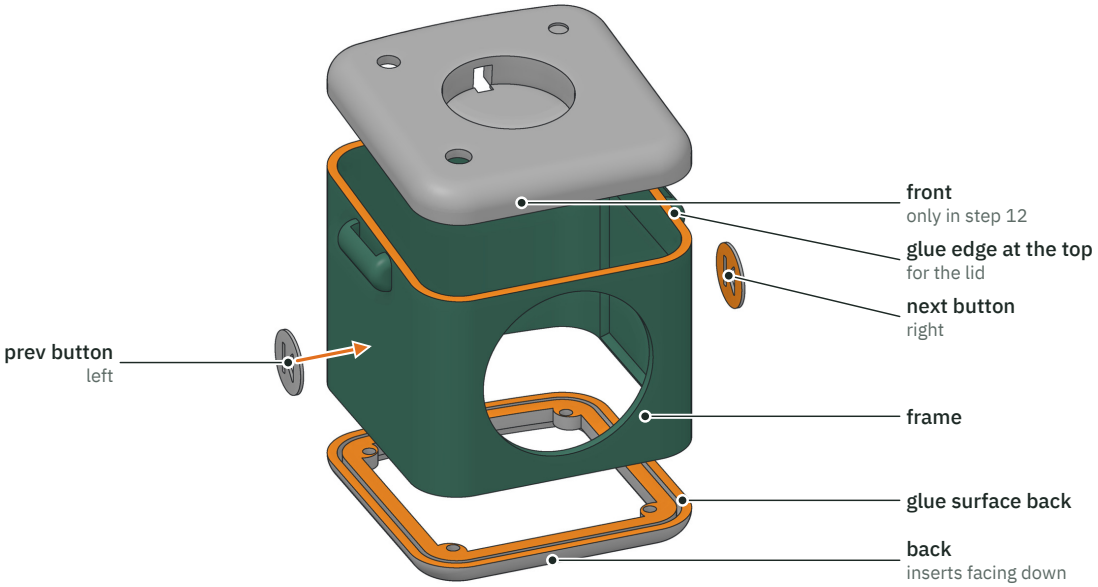



**Fig. 9** *Glue surfaces in orange. The buttons are glued to the outside of the side wall.*

### 3 Glue on the buttons

Super glue

- `prev button` onto the left, `next button` onto the right side wall, seen from the front. Both sit centred under the lug, their centre 31 mm above the bottom edge of the frame.
- Align the symbol horizontally. Apply glue sparingly, it must not ooze out of the symbol's cutouts.

### 4 Glue the bottom to the frame

Super glue

- `back` with the inserts facing down against the bottom edge of `frame`. The curved side goes to the back, towards the USB-C socket.
- Glue them on a flat surface and align the edges flush. Do not glue on the lid yet.

## 6.3 Walls

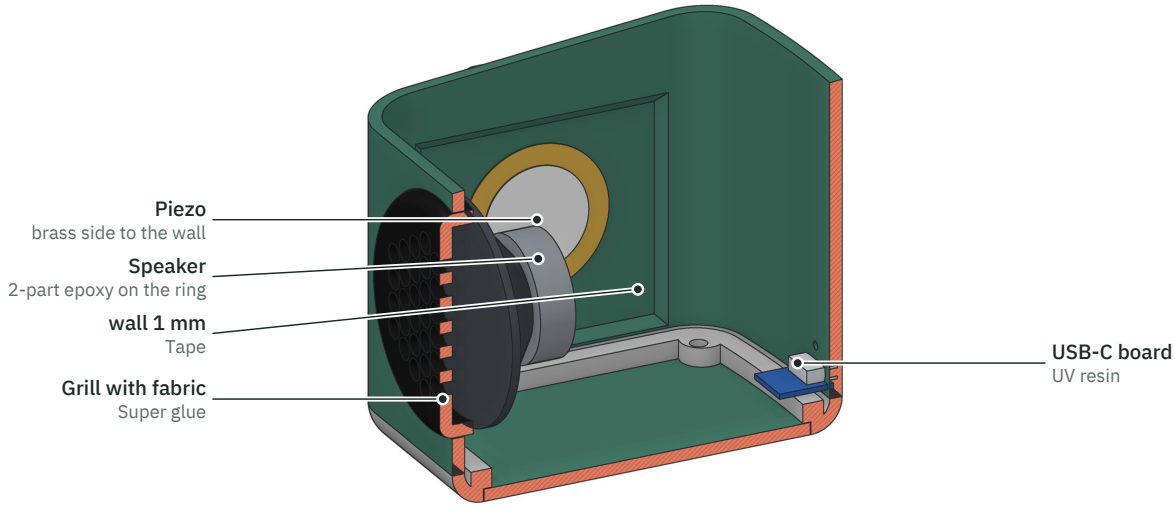



**Fig. 10** *Frame in section: what sits on the inside of the walls.*

### 5 Cover the speaker grill with fabric

Super glue

- Pull a piece of thin fabric, a sock for example, tight over the outside of the grill.
- Press the grill into the opening from the outside. The fabric gets clamped in the gap between rim (Ø 54.6 mm) and opening (Ø 55 mm).
- Fix it from the inside with super glue and cut off the excess fabric on the inside.

> **Tip:** Only apply glue on the inside. Super glue soaks into fabric, hardens and leaves pale stains.

### 6 Glue in the speaker

2-part epoxy

- Solder the wires on first, long enough to reach the main board under the lid.
- Glue the speaker onto the grill from the inside with 2-part epoxy. At 50–57 mm diameter its rim rests on the ring on the inside of the grill, a smaller one sits inside the ring right at the grill.
- Glue only on the rim, not on the cone. Let it cure.

### 7 Stick on the piezos

Double-sided tape

- Stick one piezo disc each with thin double-sided tape from the inside onto the thinned wall behind `prev button` and `next button`, centred behind the button, brass side to the wall.
- Label the wires: left is “previous” (J2), right is “next” (J3), see chapter 5.4.

> **Tip:** Thin, firm tape transfers the press better than thick foam tape. The thresholds get tuned in step 12.

## 8 Glue in the USB-C socket

UV resin

- Solder the wires on first: VBUS and GND to the TP4056, D+ and D– to the cable to the main board (chapter 5.4).
- Put the board into the cutout at the bottom of the back wall from the inside, the socket flush with the outside.
- Plug in a USB-C cable so the socket sits straight. Then fix it from the inside with UV resin and cure it. No resin in the socket.

## 6.4 Lid

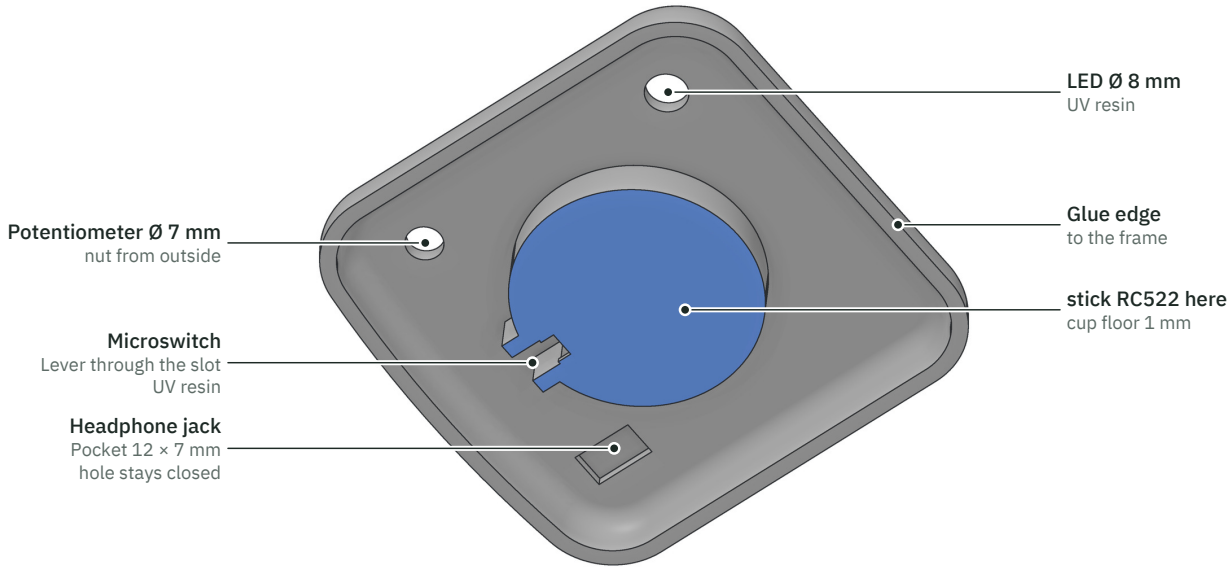



Fig. 11 `front` *from the inside. Blue: area for the RC522.*

## 9 Fit out the lid

UV resin Double-sided tape

- **LED:** Push the PL9823 from the inside up to its collar into the 8 mm hole at the front left and fix it with UV resin.
- **Potentiometer:** Push it from the inside through the 7 mm hole at the back left, washer and nut on the outside. Break off the anti-rotation tab on the potentiometer if there is one; the lid has no hole for it. Cut the shaft so that it sticks out 8.5 mm above the lid surface, then the knob sits with 0.5 mm clearance. Push on the knob `volume`.
- **Microswitch:** Put it into the pocket at the back of the token cup so that the lever reaches through the slot into the cup. Check that a placed token presses it audibly, then fix it with UV resin and cure. Resin only on the switch body, not on lever and plunger.
- **RC522:** Stick it under the token cup from the inside with thin double-sided tape, flat back facing up, antenna centred under the cup. The pin header points down, that is where the main board plugs in later.
- **Headphone jack:** Stays empty for now, the hole at the back right stays closed. There is a pocket on the inside for a later jack; then drill through the 0.2 mm layer with 6 mm.

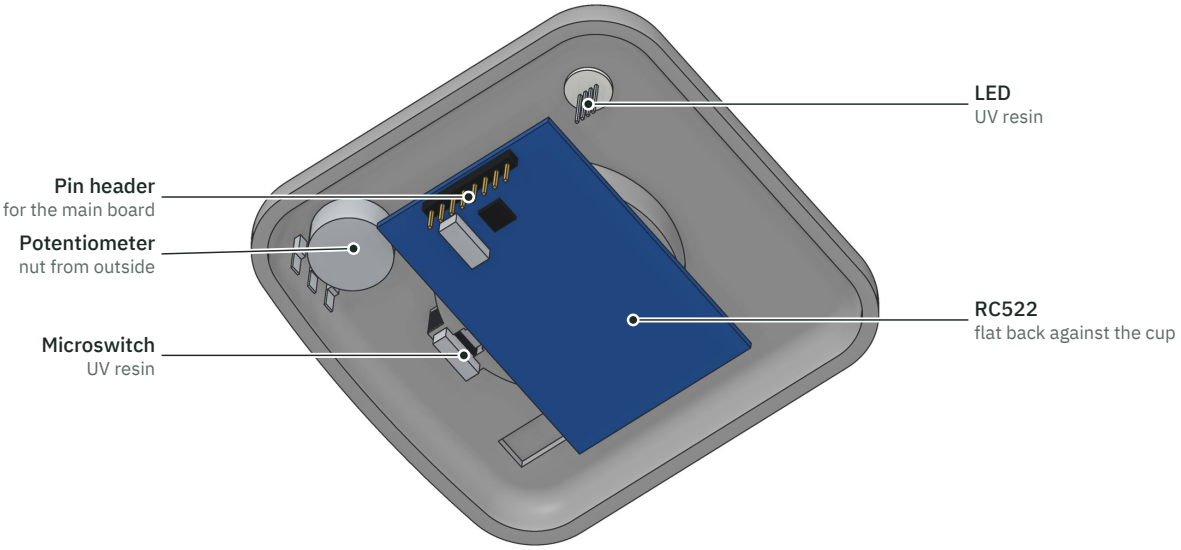



**Fig. 12** *Lid after step 9. The main board goes onto the RC522's pin header in step 11.*

## 6.5 Electronics and cells

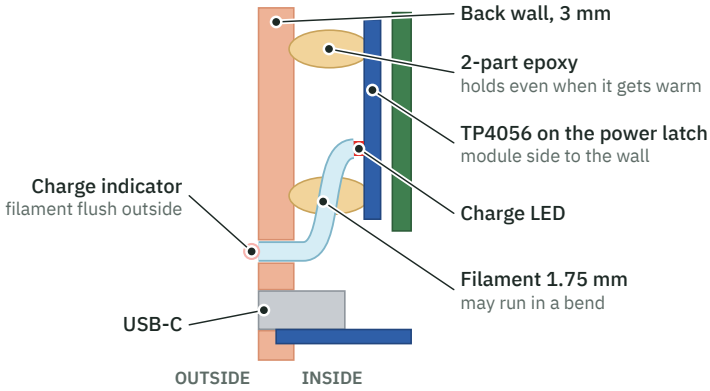



**Fig. 13** *Section through the back wall at the USB-C socket: the filament carries the light of the charge LED to the outside and may run in a bend.*

## 10 Glue in the charging board and the step-up

2-part epoxy

- Glue the charging board, that is the power latch with the TP4056 module on its back, to the inside of the wall with 2-part epoxy, close to the USB-C socket. Hot glue is not enough here: the charge controller gets very warm while charging, and hot glue can come loose then.
- Use fast-curing 2-part epoxy and hold the board until it grabs. No glue on the charge LED.
- Push a piece of clear filament (1.75 mm) from the outside through the 2 mm hole above the USB-C socket and guide it on the inside up to the charge LED. It may be longer and run in a gentle bend, so you are free in placing the charging board. Cut it flush on the outside. If it does not hold by itself, secure it with a drop of UV resin.
- Glue the step-up converter to a free spot on the inside of the wall with 2-part epoxy as well, not on the thinned button areas. It gets warm under load too.

> **Tip:** While charging, the filament next to the USB-C socket glows in the colour of the LED: depending on the module red while charging, blue or green when the battery is full.

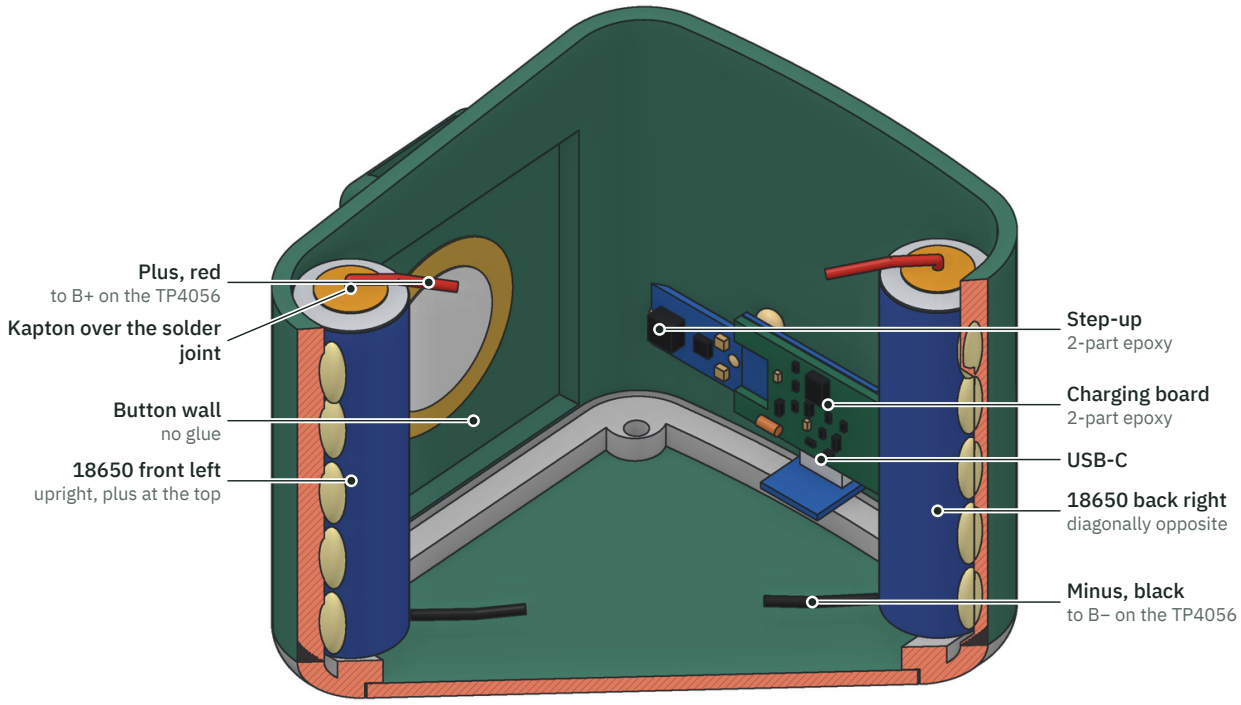



**Fig. 14** *Cells upright in two diagonally opposite corners, here front left and back right. On the back wall the charging board and the step-up. Frame cut open, lid not glued on yet.*

## 11 Wire up the electronics and glue in the cells

Electronics from chapter 5 · Hot glue

- Plug the main board from below onto the RC522's pin header and clip the spacer in at the free end (chapter 5.2, step E6).
- To be safe, put a cable tie around main board, spacer and RC522, at the end opposite the pin header. There is room between RC522 and lid there, because the token cup only carries the RC522 in the middle.
- Wire everything following the tables in chapter 5.4: cable to J4, speaker to the MAX98357A, piezos to J2 and J3, potentiometer to J1, LED to J5, microswitch with C to Trigger + and NO to Trigger In of the latch.
- Stand the two 18650 cells upright in two diagonally opposite corners, plus at the top, for example front left and back right. At the back left the potentiometer sits right above.
- The cells stand on the bottom right above the heat-set inserts and reach to just below the lid. Plain cells without protection circuit (65 mm) fit, the protection circuit sits on the TP4056 module.
- Fix them with hot glue in the gaps between cell and corner. No glue on the thinned wall areas of the buttons, it would damp the piezos.
- Wire them in parallel, plus to plus and minus to minus, and connect them to B+ and B– of the TP4056. Insulate all solder joints with Kapton tape or heat-shrink tubing.

> **Caution:** Before connecting them, bring both cells to the same voltage (at most about 0.1 V difference), otherwise a high equalising current flows. Only use identical cells, take cells with solder tabs and do not solder to the cell directly. Nothing bare may touch screws, inserts or other parts.

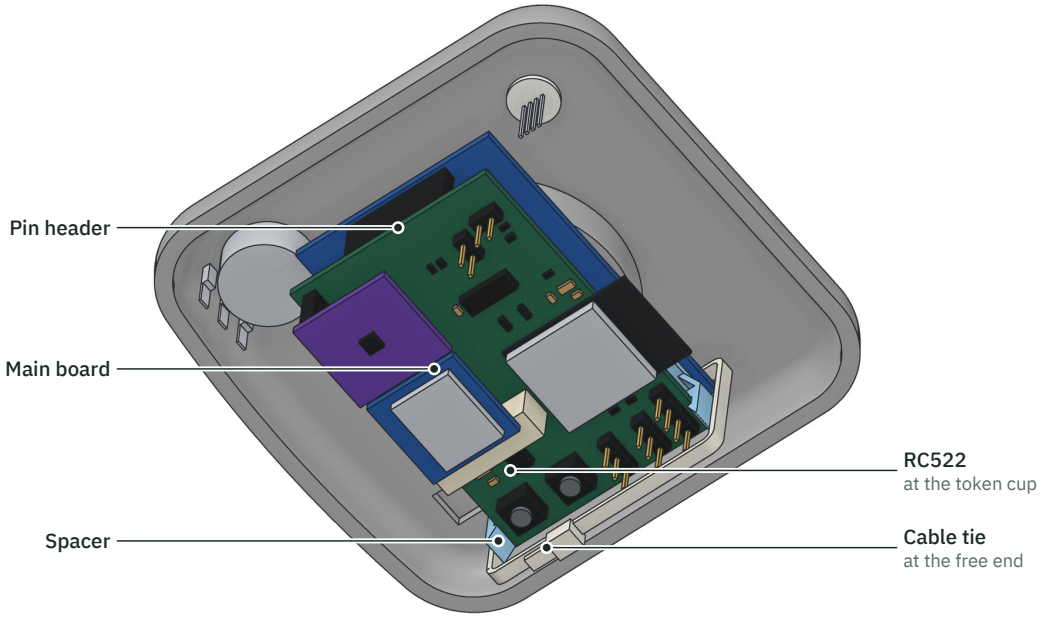



**Fig. 15** *Lid with electronics from the inside. Opposite the pin header, spacer and cable tie hold the main board.*

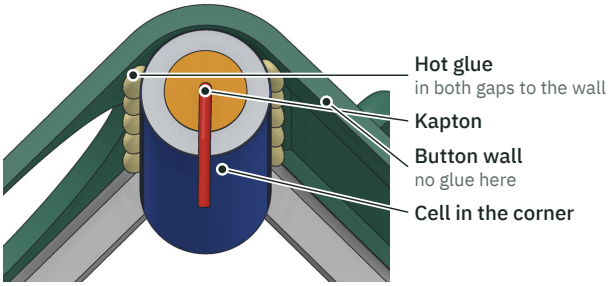



**Fig. 16** *Front left corner from above.*

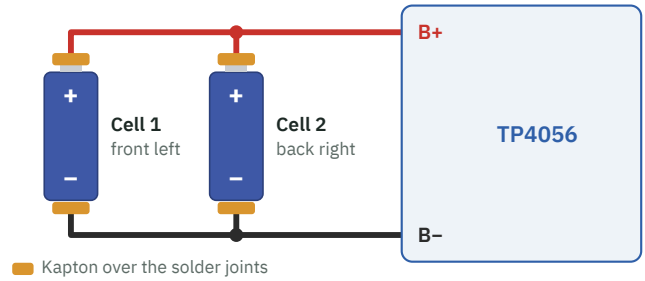



**Fig. 17** *Parallel wiring: plus to plus, minus to minus.*

## 6.6 Finishing

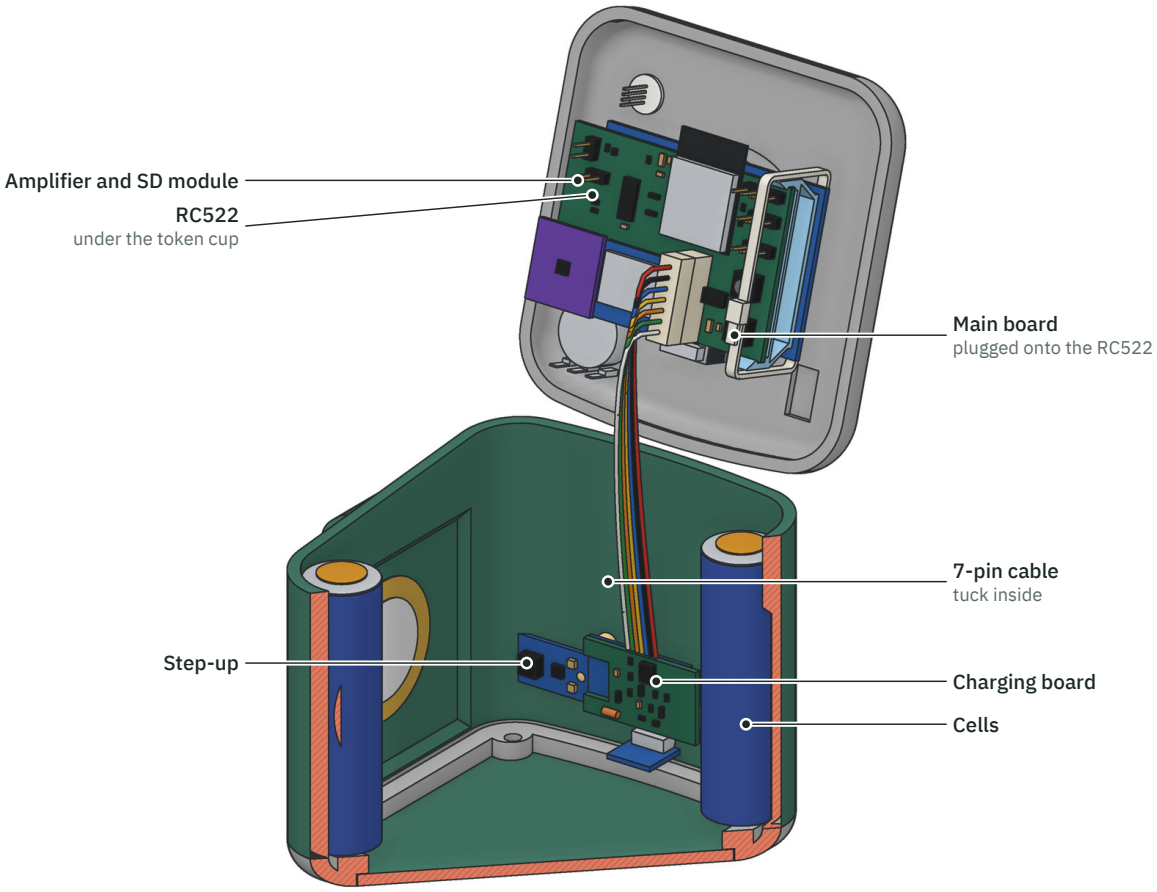



**Fig. 18** *Before gluing: the lid carries the RC522, the main board with its modules, LED, potentiometer and microswitch. The 7-pin cable connects it to the power side.*

### 12 Test, then glue on the lid

Super glue

- Start it up before gluing, as described in chapter 5.5: place a token, box starts, flash, check volume, LED and battery gauge.
- Tune the knock sensors with `knockdebug:1` and `knockdebug:2` (doc/development.md). The thresholds are set per box in `platformio.ini`, a new box gets its own env there.
- Tuck the wires inside, nothing may lie on the glue edge. Glue the lid onto the frame with super glue, the curved side to the back. Then quickly re-check the knock sensors.

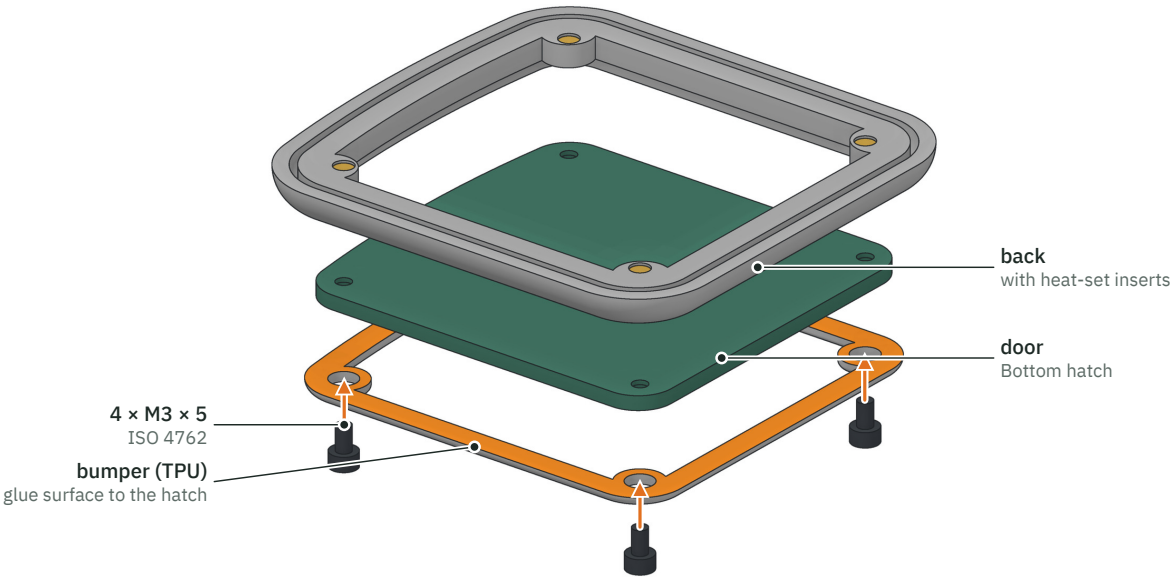



**Fig. 19** *Bottom hatch with bumper and screws. Orange: glue surface.*

## 13 Screw on the bottom hatch

Super glue

- Glue the bumper under the hatch with super glue. To align it, push two screws through bumper and hatch.
- Screw the hatch on with four M3 × 5 screws. The heads sink into the bumper. Through the hatch you can reach the electronics again.

> **Caution:** Do not use screws longer than M3 × 5. In two corners a cell sits right above the insert.

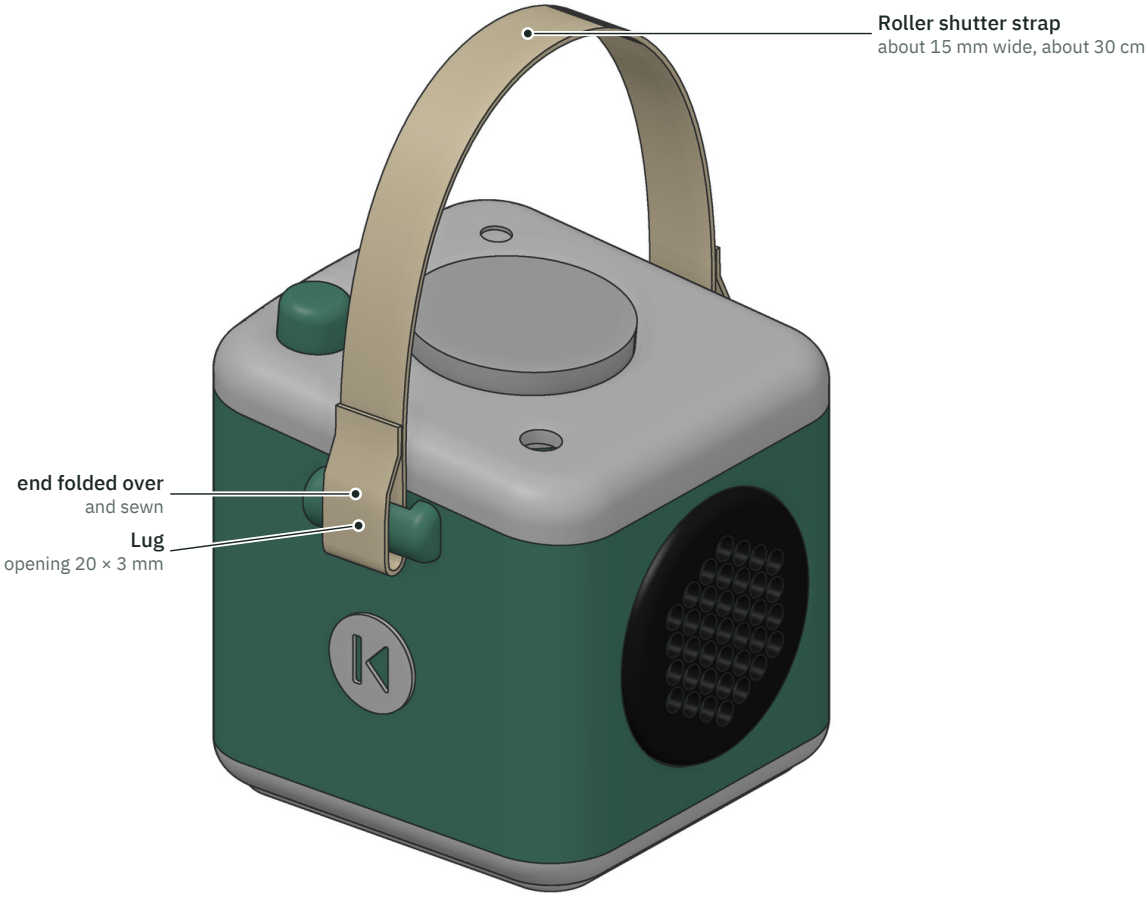



**Fig. 20** *For carrying: a piece of roller shutter strap through both lugs.*

## 14 Fit the carrying strap

Needle and thread

- Take about 30 cm of roller shutter strap from the hardware store, about 15 mm wide.
- Pass each end through a lug, fold it over and sew it down, for example with a box stitch with a cross.
- Now the box can be carried by the strap.

> **Tip:** Briefly melt the cut edges with a lighter, then the strap does not fray.

# 7 Dimensions and interfaces

All values from the STEP model, in millimetres. Paths refer to the repo github.com/aaronblankenburg/timon-box.

| Location | Dimension |
|---|---|
| Outside | 94 × 90 mm, 102 mm wide with lugs; height with bumper 94.5 mm at the back, 87 mm at the front |
| Lid | tilted forward by 5°, 3 mm top plate, 8 mm rim |
| Walls | 3 mm, thinned to 1 mm over an area of 56 × 50 mm at the buttons |
| Lugs | opening 20 × 3 mm, for straps up to just under 20 mm wide |
| Token cup | Ø 40 mm, 10 mm deep, floor 1 mm |
| Token | body Ø 39.7 mm, rim Ø 44 mm, 15 mm tall; NFC pocket Ø 26 × 0.4 mm, 0.2 mm above the bottom face |
| LED | hole Ø 8 mm |
| Potentiometer and knob | hole Ø 7 mm; knob Ø 14 × 9 mm, shaft hole Ø 6.2 mm (8 mm deep from the bottom edge), recess Ø 12 × 3 mm |
| Headphone jack | hole Ø 6 mm, closed at the top by 0.2 mm; pocket 12 × 7 × 2 mm |
| USB-C | cutout 9.2 × 3.5 mm, corners R 1.5; 5 mm above its centre a Ø 2 mm hole for the light pipe |
| Speaker | opening Ø 55 mm; grill with rim Ø 54.6 mm, seating ring Ø 48.6–54.6 mm, protrudes 3 mm |
| Buttons | Ø 20 × 1 mm, centre 31 mm above the bottom edge of the frame |
| Heat-set inserts | 4 holes Ø 4.0 mm, 4.5 mm deep, through |
| Hatch | 2.9 mm thick, counterbore Ø 5.6 × 2 mm, clearance hole Ø 3.1 mm |
| Bumper | 2 mm, TPU |
| Spacer | 44.2 × 16.8 × 10 mm; 41.2 mm wide between the arms, snap hooks at 39.9 mm |
| Board stack | about 11 mm between main board and RC522: female header 8.5 mm and the plastic of the pin header 2.5 mm |

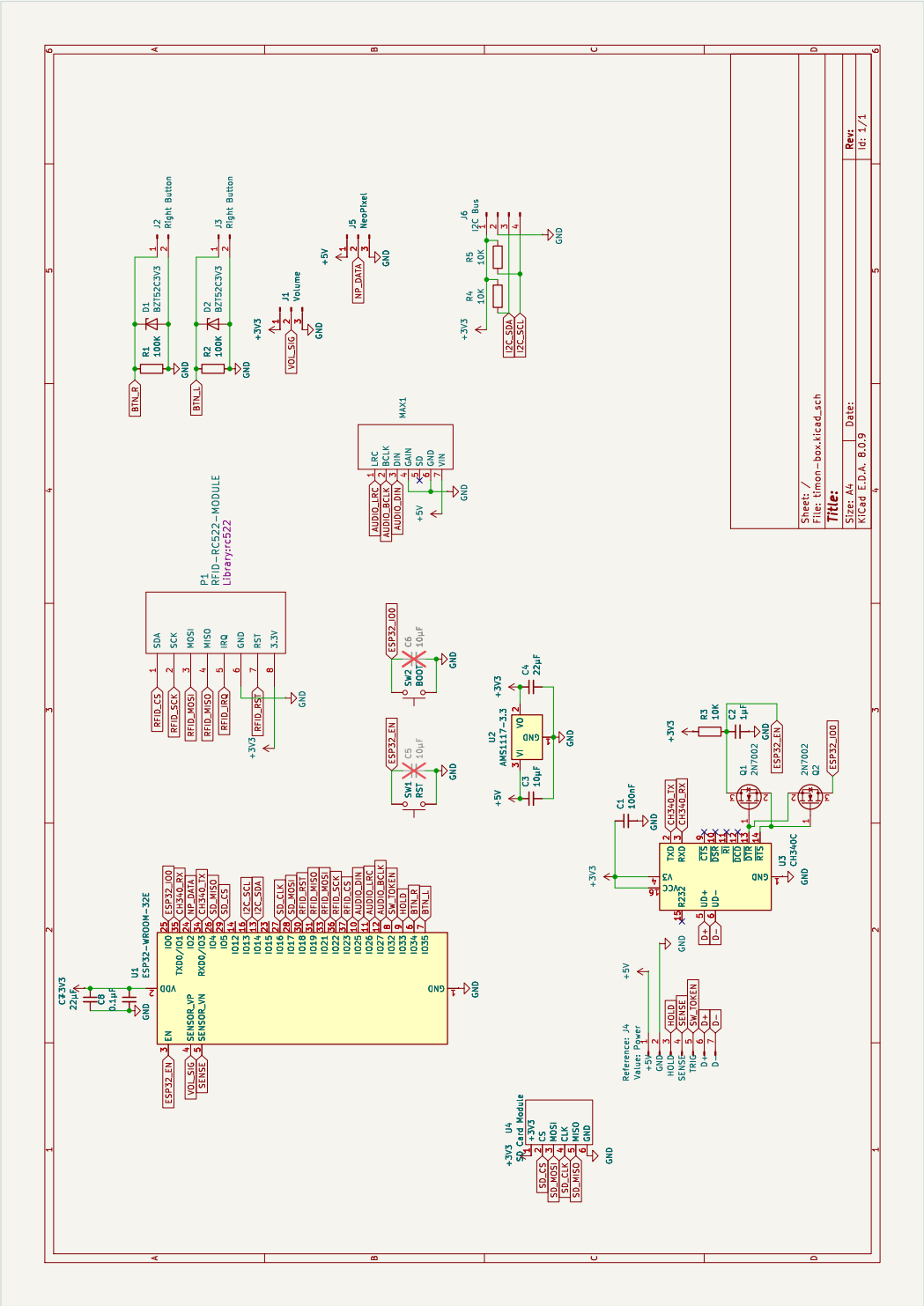
## Files

| Path | Contents |
|---|---|
| `models/box.step` | CAD model, all ten parts in assembled position |
| `models/pcb_spacer.step` | Spacer between main board and RC522 |
| `models/plates/` | Ready-made print plates as 3MF, one per colour, and a Bambu Studio project with all of them |
| `models/stl/` | Every printed part as STL, in print orientation |
| `hardware/` | KiCad projects, schematics and Gerbers of the boards, wiring of the power side (WireViz) |
| `doc/hardware.md` | Electronics: parts list, boards, wiring, first start |
| `README.md` | Overview and quick start |
| `doc/usage.md` | How to use the box |
| `doc/development.md` | Tuning the knock sensors, development |
| `tools/token-sheet.html` | Token pictures as a 10 × 15 cm photo print |
| `doc/assembly/` | Sources of this guide: text, images from the model, build script |

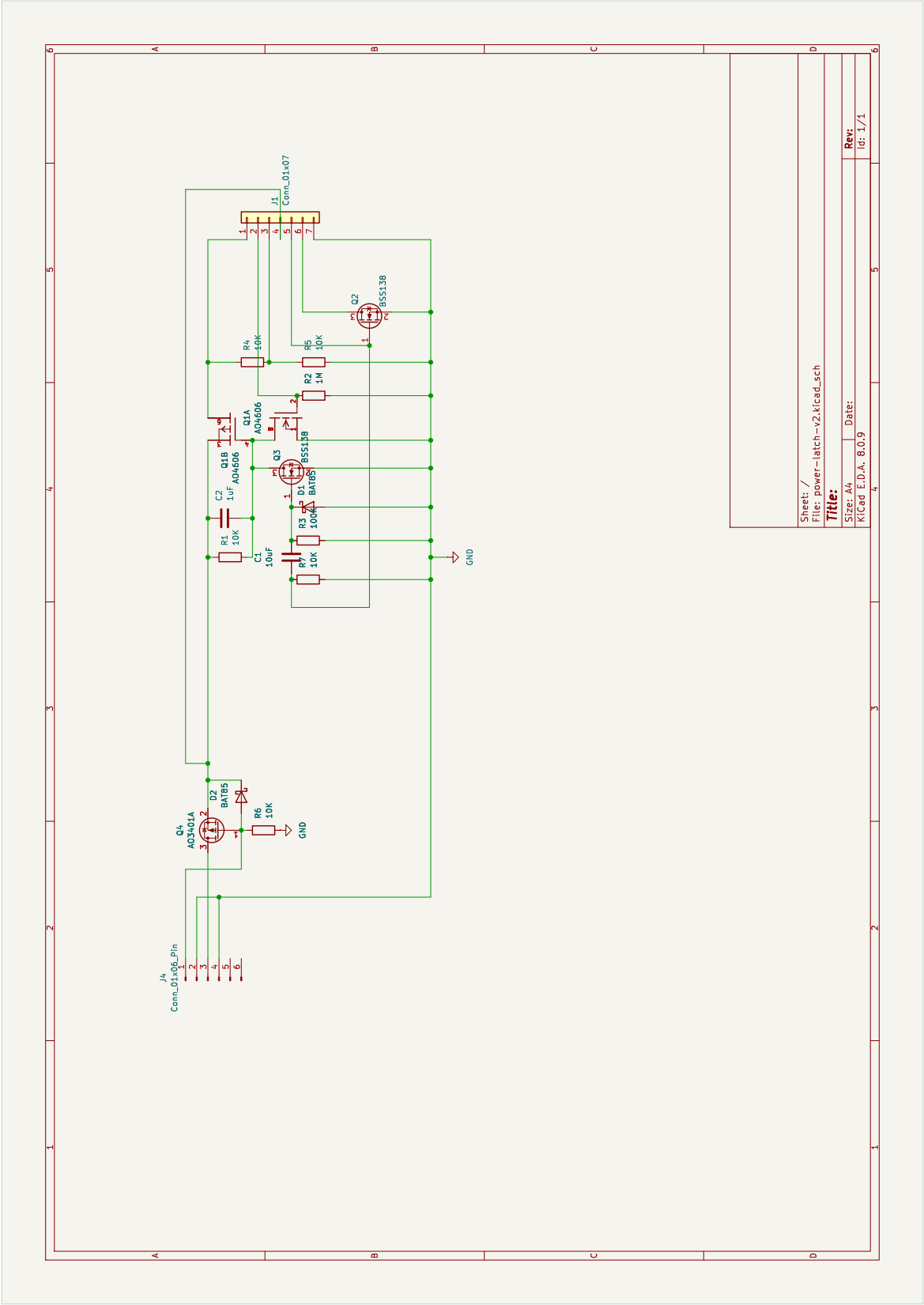
# 8 Appendix

Schematics and wiring from `hardware/`, included unchanged. On screen they can be zoomed in as far as needed.

## Schematic of the main board



## Schematic of the power latch V2



## Wiring of the power side

Generated from `hardware/wiring/power-side.yml` with WireViz. Every row in a cable names both ends.

